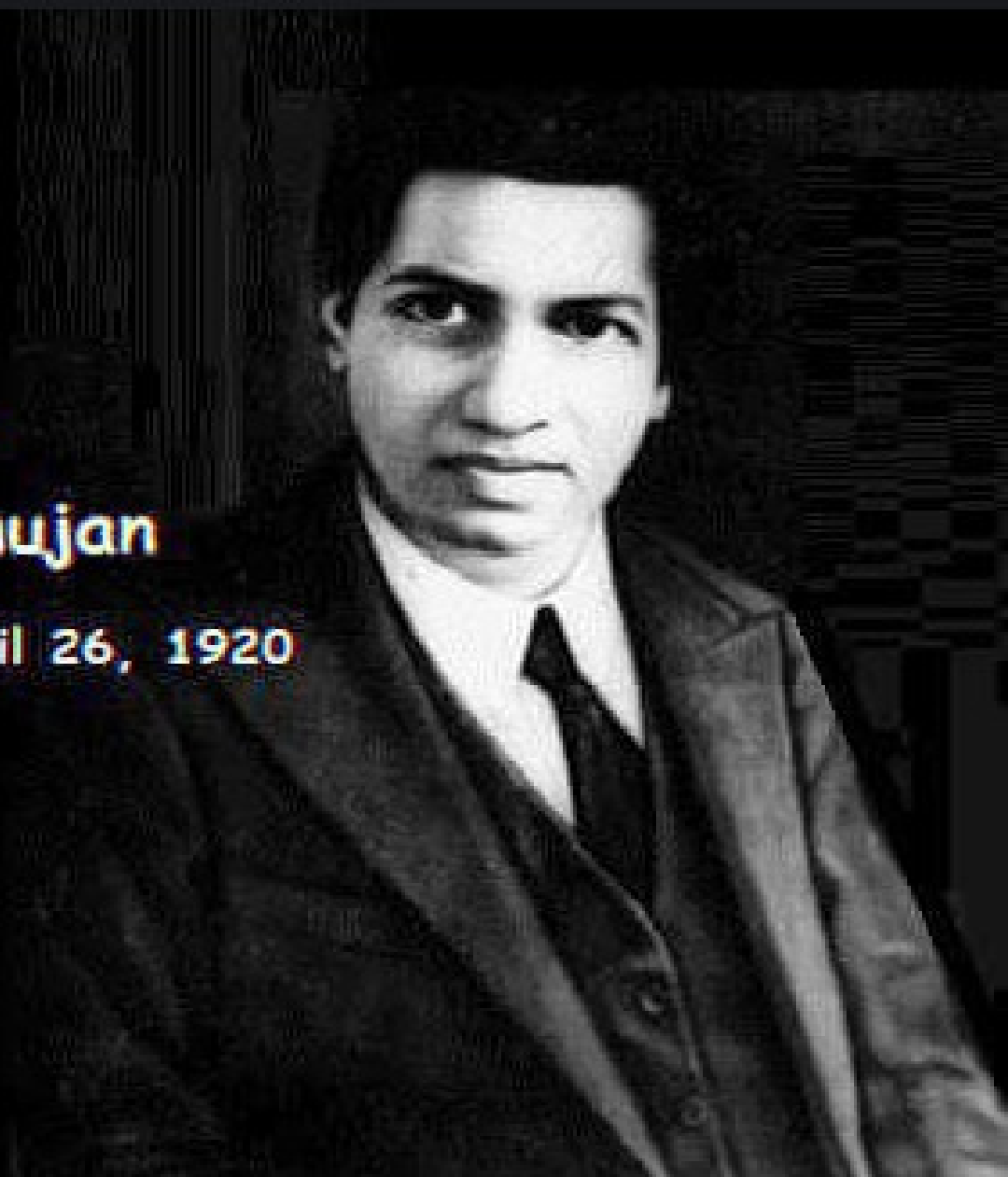


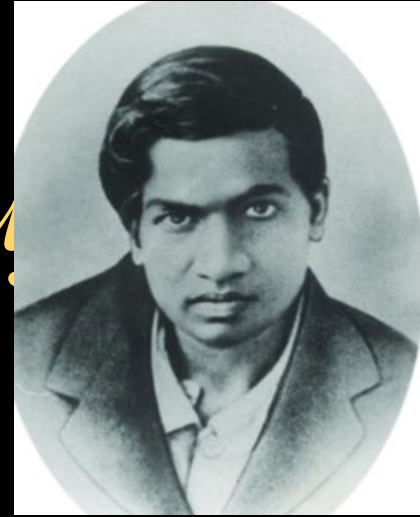
Mathematician

Srinivasa Ramanujan

December 22, 1887 – April 26, 1920



Srinivasa Ramanujan



- **Born** : 22 December 1887
Erode
- **Other names** : Srinivasa Ramanujan
Aiyangar
- **Marriage** : July 14, 1909
- **Wife** : Janakiammal
- **Died** : 26 April 1920 (aged 32)
Kumbakonam

EDUCATION

- **Government Arts College** (no degree)

- At age 15 Srinivasa Ramanujan obtained a mathematics book containing thousands of theorems, which he verified and from which he developed his own ideas.

- In 1903 he briefly attended the **University of Madras**.

- In 1913 he began a **postal partnership** with the English mathematician G. H. Hardy at the **University of Cambridge, England**

- In 1914 he went to England to study at **Trinity College, Cambridge**, with British mathematician **G.H. Hardy**. (Bachelor of Arts by Research, 1916)



CONTRIBUTIONS





Ramanujan's nested radical

$$\sqrt{1+2\sqrt{1+3\sqrt{1+4\sqrt{1+\dots}}}}$$

He then formulated an equation to solve the infinitely nested radical problem.

$$x + n + a = \sqrt{ax + (n + a)^2 + x \sqrt{a(x + n) + (n + a)^2 + \dots}}$$

Ramanujan's nested radical



S. Ramanujan (1887-1920)

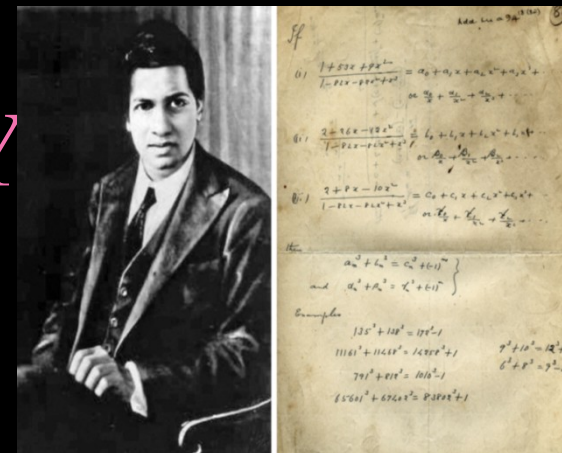
$$3 = \sqrt{1 + 2\sqrt{1 + 3\sqrt{1 + 4\sqrt{1 + \dots}}}}$$

Proof: Define $f(x) = x + n + a$, so that $f(x)^2 = ax + (n + a)^2 + xf(x + n)$.

Set $a = 0$, $n = 1$, $x = 2$ and substitute recursively for $f(x)$.



RAMANUJAN- HARDY NUMBER



- The Hardy-Ramanujan number is named such after an anecdote of the British mathematician G.H.Hardy who had gone to visit Ramanujan in hospital in a taxi cab numbered 1729.
- It is also known as the Taxicab number.
The smallest number that can be expressed as the sum of two cubes in two distinct ways.
- 1729 is the sum of the cubes of 10 and 9 and also the sum of the cubes of 12 and 1.
 $1729 = 1^3 + 12^3 = 9^3 + 10^3$.

Contribution to the theory of Partition.

A partition of a positive integer ' n ' is a non-increasing sequence of positive integers, whose sum equals n .

- Generally, it means the number of ways in which a given number can be expressed as a sum of positive integers.
- For example, $p(4) = 5$, i.e. there are five different ways that we can express the number 4. The partitions of the number 4 are:

4,

3+1,

2+2,

2+1+1,

1+1+1+1.

Number n	Number of partitions p(n)
1	1
2	2 = {2, 1+1}
3	3 = {3, 1+2, 1+1+1}
4	5 = {4, 3+1, 2+2, 1+2+1, 1+1+1+1}
5	7
6	11
7	15
8	22
9	30
10	42
100	190,569,292
200	3,972,999,029,388



interesting

RAMANUJAN'S MAGIC

SQUARE

22	12	18	87
88	17	9	25
10	24	89	16
19	86	23	11

Sum of any row is 139

22	12	18	87
88	17	9	25
10	24	89	16
19	86	23	11

Sum of any column is 139

22	12	18	87
88	17	9	25
10	24	89	16
19	86	23	11

Sum of the numbers of any diagonal is also 139

22	12	18	87
88	17	9	25
10	24	89	16
19	86	23	11

Sum of the numbers of corner cells is 139

22	12	18	87
88	17	9	25
10	24	89	16
19	86	23	11

Sum of the
identical
colored
boxes are
139

22	12	18	87
88	17	9	25
10	24	89	16
19	86	23	11

22	12	18	87
88	17	9	25
10	24	89	16
19	86	23	11

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22	12	18	87
88	17	9	25
10	24	89	16
19	86	23	11

22. 12 .1887

Numbers in the first row is the Date of Birth of Ramanujan.



OTHER CONTRIBUTIONS

- Infinite series for p - Ramanujan Summation
 - (i) A technique of assigning a value to the divergent series
 - (ii) Basis of many algorithms we use today
 - Calculation of the digits in π
 - Game Theory Development
 - Have also contributed in branches like Complex Analysis, Number Theory, Continued fractions etc
- 



OTHER CONTRIBUTIONS(cont.)

- Landau–Ramanujan constant (Number Theory)
 - Mock theta functions
 - Ramanujan conjecture
 - Ramanujan prime
 - Ramanujan–Soldner constant
 - Ramanujan theta function
 - Ramanujan's sum
 - Rogers–Ramanujan identities
 - Ramanujan's master theorem
 - Ramanujan–Sato series
- 

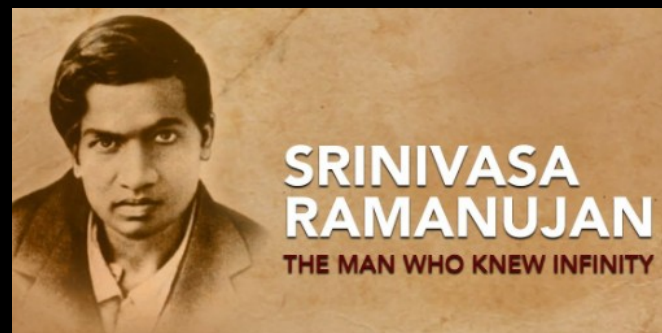


Srinivasa Ramanujan's birthday, December 22, was declared as the **National Mathematics Day**

in 2012 by the then Prime Minister Dr. Manmohan Singh.

- **The Man Who Knew Infinity** is a 2015 British biographical drama film about the Indian mathematician Srinivasa Ramanujan, based on the 1991 book of the same name by Robert Kanigel.

- Government of India first released **postal stamp** on Srinivas Ramanujan on 22nd December 1962. Later, new stamp was released on same date in 2011



Thank
you

